

A Mini Project Report

On

**Precision Agro Analytics: Mitigating Environmental Impact
through ML-Driven Crop and Soil Health Optimization**

Submitted in Partial fulfillment of requirements for the award of the
degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE & ENGINEERING

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July - 2026



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CERTIFICATE

This is to certify that this is a Bonafide record of the project report titled
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- To be the fountain head in producing highly skilled, globally competent engineers. Producing quality graduates trained in the latest software technologies and related tools and striving to make India a world leader in software products and services.

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PROGRAM OUTCOMES (POs)

PO1. Engineering Knowledge: Apply the knowledge of mathematics, science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem Analysis: Identify formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences

PO3. Design/Development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct Investigations of Complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

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PO6. The Engineer and Society: Apply reasoning informed by contextual knowledge to societal, health, safety. Legal und cultural issues and the consequent responsibilities relevant to professional engineering practice.

PO7. Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1. An ability to analyze the common business functions to design and develop appropriate Information Technology solutions for social upliftment.

PSO2. Shall have expertise on the evolving technologies like Mobile Apps, CRM, ERP, Big Data, etc.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO1. Graduates will have successful careers in computer related engineering fields or will be able to successfully pursue advanced higher education degrees.

PEO2. Graduates will try and provide solutions to challenging problems in their profession by applying computer engineering principles.

PEO3. Graduates will engage in life-long learning and professional development by rapidly adapting to changing work environment.

PEO4. Graduates will communicate effectively, work collaboratively and exhibit high levels of professionalism and ethical responsibility.

PROJECT OUTCOMES

P1: Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program to the real-world problems.

P2: Evaluate different solutions based on economic and technical feasibility

P3: Effectively plan a project and confidently perform all aspects of project management.

P4: Demonstrate effective coding, written, presentation and oral communication skills.

MAPPING PROJECT OUTCOMES WITH PROGRAM OUTCOMES

PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
P1	3	3	2	2	1	-	-	-	-	-	-
P2	2	3	3	2	1	1	1	-	-	-	2
P3	-	1	2	1	1	-	-	-	3	2	3
P4	-	-	-	-	1	-	-	1	2	3	1

1– LOW

2 –MEDIUM

3– HIGH

**PROJECT OUTCOMES MAPPING WITH PROGRAM SPECIFIC
OUTCOMES**

PSO	PSO1	PSO2
P1	3	2
P2	3	2
P3	2	1
P4	1	1

**PROJECT OUTCOMES MAPPING WITH PROGRAM EDUCATIONAL
OBJECTIVES**

PEO	PEO1	PEO2	PEO3	PEO4
P1	3	3	2	1
P2	2	3	2	1
P3	2	2	2	3
P4	2	1	2	3

DECLARATION

We hereby declare that the results embodied in the document entitled "**Precision Agro Analytics: Mitigating Environmental Impact through ML-Driven Crop and Soil Health Optimization**" has been carried out by us together during the academic year 2025-26 as a fulfillment for the completion of BE III year VI Semester Course Work of the B.E degree in Computer Science & Engineering from Osmania University. We have not submitted this document to any other university or organization for the award of any other degree.

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We are also thankful to **Dr. K.V.S Sudhakar**, Project Coordinator, for supporting us from project selection till the submission to complete this project within the scheduled time.

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ABSTRACT

The integration of machine learning into agriculture is a transformative development for sustainable food security. Unexpected frost events are disrupting traditional farming schedules that farmers have relied on for generations. To help address this problem, this research presents a comprehensive web-based platform designed to optimize crop selection and soil management through data-driven insights. Utilizing a MERN-stack architecture, the system allows users to input critical environmental and soil parameters including fertilizer measurements, rainfall, temperature, and soil pH. Advanced machine learning algorithms such as Random Forest are employed to process these inputs, delivering an output consisting: a ranked recommendation of the top five suitable crops, a quantitative assessment of soil health, and tailored fertilizer suggestions. By shifting from generalized regional data to site-specific analysis, this system empowers farmers with real-time, actionable intelligence, ultimately improving resource efficiency and reducing the environmental footprint of traditional farming practices. The system bridges the gap between data science and real world farming needs, supporting farmer livelihoods, and working toward a more sustainable future for agriculture worldwide.

By leveraging machine learning-driven insights, the system promotes precision agriculture practices, enabling optimal utilization of water, fertilizers, and other resources. Furthermore, the platform can be continuously improved through the integration of new datasets, making it adaptable to changing climatic conditions and evolving agricultural requirements. This contributes to increased crop productivity, economic sustainability, and long-term agricultural resilience. Ultimately, this research demonstrates the potential of machine learning and data-driven technologies to revolutionize modern agriculture by enabling smarter, more sustainable farming practices. The proposed system enhances agricultural productivity and environmental sustainability.

Keywords: Agriculture, Food Security, Machine Learning, MERN Stack, Crop Recommendation, Random Forest, Fertilizer Optimization, Soil Health.

LIST OF FIGURES

Fig. No.	Name of the Figure	Page No.
1.	Architecture diagram	3
2.	Use Case Diagram	9
3.	Class Diagram	10
4.	Sequence Diagram	11
5.	Activity Diagram	12
6.	State Chart Diagram	13
7.	Deployment Diagram	14

LIST OF TABLES

Table No.	Name of the Table	Page No.
1.	Software requirements	7
2.	Hardware requirements	8

CONTENTS

Abstract	i
List of Figures	ii
List of Tables	iii
CHAPTER-1 INTRODUCTION	1
1.1 Purpose of the Project	
1.2 Scope of the Project	
1.3 Problem with existing systems	
1.4 Architecture Diagram	
CHAPTER-2 LITERATURE SURVERY	4
2.1 Overview	
2.2 Papers	
2.2.1 Identification of Fertilizer Based NPL Values and Crop Recommendation Using ML Algorithm	
2.2.2 Developing fertilizer recommendations for rice in Sub-Saharan Africa, achievements and opportunities	
2.2.3 Intelligent Crop Selection: Leveraging Machine Learning for Soil-Informed Agriculture	
2.2.4 Crop Prediction using Machine Learning Approaches	
CHAPTER-3 SOFTWARE REQUIREMENT SPECIFICATION	6
3.1 Introduction to SRS	
3.2 Functional Requirements	
3.3 Non-Functional Requirements	
3.4 Software Requirements	
3.5 Hardware Requirements	
CHAPTER-4 SYSTEM DESIGN	9
4.1 Introduction to UML	
4.2 UML Diagrams	
4.2.1 Use Case Diagram	
4.2.2 Class Diagram	
4.2.3 Sequence Diagram	
4.2.4 Activity Diagram	

4.2.5 State Chart Diagram

4.2.6 Deployment Diagram

CHAPTER-5	IMPLEMENTATION	15
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5.1	Include sample code and technologies description	
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CHAPTER-6	RESULTS	18
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6.1	Include colour-printed screenshots of outputs, results, reports, etc	
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CHAPTER-7	CONCLUSION	20
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7.1	Summarize project findings	
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REFERENCES

CHAPTER 1

INTRODUCTION

1.1 Purpose of the Project

The primary purpose of this project is to assist farmers in making informed agricultural decisions by leveraging machine learning and data-driven analysis. Traditional farming methods often rely on experience, seasonal patterns, and generalized regional recommendations, which may not be effective under changing climatic conditions and varying soil characteristics. This project aims to overcome these limitations by providing personalized crop recommendations based on real-time environmental and soil parameters such as rainfall, temperature, soil pH, and fertilizer measurements. By analyzing these factors using machine learning algorithms like Random Forest, the system identifies the most suitable crops for a specific location, helping farmers maximize productivity and reduce the risk of crop failure.

Another important objective of the project is to promote sustainable and efficient resource management in agriculture. In addition to recommending crops, the platform evaluates soil health and provides customized fertilizer suggestions, enabling farmers to maintain soil fertility while avoiding excessive use of agricultural inputs. Through its user-friendly web-based interface developed using the MERN stack, the system makes advanced agricultural insights accessible to users with minimal technical expertise. Ultimately, the project seeks to bridge the gap between modern technology and practical farming needs, supporting higher crop yields, improved economic outcomes for farmers, and environmentally responsible agricultural practices.

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1.2 Scope of the Project

This project focuses on the development of a machine learning based web platform that assists farmers in selecting suitable crops and managing soil health through data-driven recommendations. The system utilizes key agricultural parameters such as rainfall, temperature, soil pH, and fertilizer measurements to analyze field conditions and generate personalized recommendations. By employing machine learning algorithms, particularly Random Forest, the platform predicts the top five crops best suited for the given environmental conditions while also providing soil health assessments and fertilizer suggestions. The web based implementation using the MERN stack ensures accessibility, scalability, and ease of use for farmers.

The scope of the project extends to supporting precision agriculture by enabling site specific decision-making rather than relying on generalized regional guidelines. The platform can help optimize resource utilization, reduce unnecessary fertilizer usage, improve crop productivity, and promote sustainable farming practices. As future enhancements, the system can incorporate a market price prediction module to help farmers make informed decisions not only about which crops to cultivate but also about the optimal time to sell them for maximum profit. The platform can also be extended to provide fertilizer and pesticide recommendations along with cost analysis, enabling farmers to balance productivity and expenditure effectively. Furthermore, the project can evolve into a comprehensive smart farming ecosystem by integrating automated irrigation management, crop disease detection, and real-time monitoring capabilities.

1.3 Problems with Existing Systems

Although several machine learning-based crop and fertilizer recommendation systems have been developed, many existing solutions still suffer from limitations that reduce their effectiveness in real-world agricultural environments. Most current systems focus on generalized recommendations and often fail to address the specific needs of individual farms. The analysis of the literature and research gaps reveals several challenges that need to be overcome to achieve accurate, sustainable, and farmer-friendly agricultural decision support systems

- **Dependence on generalized regional data:** Most existing crop recommendation systems rely on regional datasets or satellite imagery to generate recommendations. Such approaches often ignore the variations in soil composition, nutrient levels, and microclimatic conditions that exist within individual farms, leading to less accurate recommendations.
- **Lack of site specific recommendations:** Many current systems fail to provide personalized recommendations based on the exact conditions of a farmer's land. As a result, farmers may receive suggestions that are not fully suitable for their soil characteristics, weather conditions, or available resources, reducing overall productivity.
- **Inadequate fertilizer optimization and resource management:** Several existing systems focus mainly on crop prediction while providing limited guidance on fertilizer usage. This can result in inefficient fertilizer application, increased cultivation costs, nutrient imbalances in the soil, and a greater environmental impact due to excessive chemical usage.
- **Isolated Output Pipelines:** A significant portion of existing systems treat crop selection and soil nutrient management as entirely separate problems rather than an integrated ecosystem. They either recommend an optimal crop without providing matching fertilizer adjustments, or they suggest fertilizer chemical ratios without validating whether that soil composition aligns with the economic viability and market demand of the target crop.

The integration of machine learning into agricultural recommendation frameworks marks a significant technological shift, existing solutions remain fundamentally limited because of the regional data, static analytics, and isolated output pipelines. To bridge the gap, this project solves the need for an integrated, real-time decision support system.

1.4 Architecture diagram

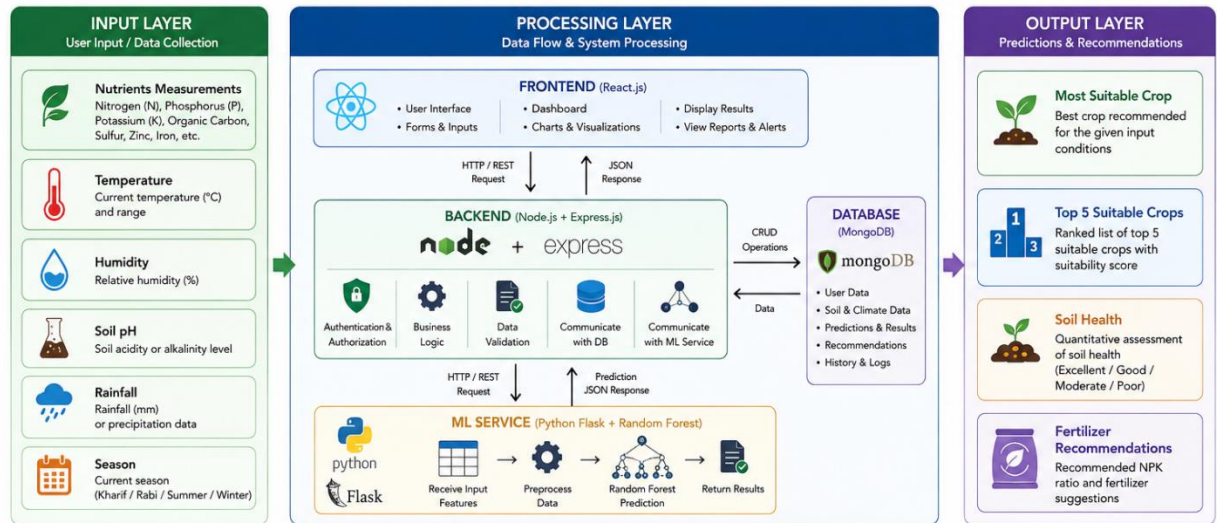


Fig 1: Architecture diagram

The proposed Precision Agro Analytics System follows a multi-layered architecture designed to provide accurate agricultural recommendations through seamless integration of web technologies and machine learning model. The architecture consists of three major layers: the Input Layer, Processing Layer, and Output Layer. The Input Layer collects critical agricultural parameters from the user, including nutrient measurements (Nitrogen, Phosphorus, Potassium), temperature, humidity, soil pH, rainfall, and season information. These parameters serve as the foundation for the recommendation process, ensuring that predictions are tailored to the specific environmental and soil conditions of a particular farm. The Processing Layer forms the core of the system and is responsible for data validation, storage, and prediction generation. User inputs are transmitted from the React.js frontend to a Node.js and Express.js backend through RESTful API requests. The backend validates the incoming data, manages business logic, handles authentication and authorization, and communicates with both the MongoDB database and the machine learning service. MongoDB stores user information, environmental data, prediction results, recommendations, and historical records for future reference. For prediction tasks, the backend sends the processed input parameters to a Python Flask-based machine learning service. The ML service performs data preprocessing and utilizes a Random Forest model trained on agricultural datasets to analyze soil and climatic conditions. After generating predictions, the results are returned to the backend in JSON format and subsequently forwarded to the frontend.

Finally, the Output Layer presents the generated insights to the user, including the most suitable crop, a ranked list of the top five suitable crops, a quantitative soil health assessment, and customized fertilizer recommendations. This modular architecture ensures scalability, maintainability, efficient data flow, and accurate decision support, making the platform a robust solution for modern precision agriculture.

CHAPTER 2

LITERATURE SURVEY

2.1 Overview

This section provides a comprehensive analysis of existing researches in crop recommendation and fertilizer management systems using machine learning. Various studies have explored algorithms such as Decision Trees, Support Vector Machines (SVM), Random Forest, Naïve Bayes, and Neural Networks to improve agricultural decision-making and crop productivity. The insights gained from the literature survey serve as the foundation for developing a more accurate, intelligent agricultural recommendation platform.

2.2 Related Approaches for Crop Recommendation

2.2.1 Identification of Fertilizer Based NPK Values and Crop Recommendation Using Machine Learning Algorithm

M. Venkatesan and co-authors proposed FertiForecast, an intelligent precision agriculture system designed to recommend suitable fertilizers and crops based on soil nutrient levels and environmental conditions. The system integrates a Raspberry Pi with multiple sensors, including NPK, pH, soil moisture, temperature, humidity, and air quality sensors, to continuously monitor field conditions. The collected data is analyzed using machine learning techniques to generate personalized fertilizer recommendations for crops such as paddy, tomato, groundnut, and chili. The study emphasizes the importance of data-driven nutrient management in improving crop productivity while reducing excessive fertilizer usage and environmental impact.

2.2.2 Developing fertilizer recommendations for rice in Sub-Saharan Africa

Vincent Boubie Bado et al. (2018) explored fertilizer recommendation techniques using Random Forest and SVM algorithms. Their findings indicated that Random Forest outperformed other models with an accuracy of approximately 95%, highlighting its suitability for agricultural recommendation systems. The study also emphasized the need for larger datasets with more diverse attributes to improve model performance.

2.2.3 Intelligent Crop Selection: Leveraging Machine Learning for Soil-Informed Agriculture

Kranti Sapkal and Avinash Kadam proposed a machine learning-based crop recommendation system that utilizes soil characteristics to assist farmers in selecting suitable crops. The study employed several machine learning algorithms including Decision Tree, Random Forest, Logistic Regression, and K-Means Clustering for crop prediction. Soil parameters such as pH, nitrogen, phosphorus, potassium, and other micronutrients were used as input features. The results showed that Random Forest achieved high prediction accuracy, demonstrating the effectiveness of machine learning techniques in providing intelligent and data-driven crop recommendations.

2.2.4 Crop Prediction Using Machine Learning Approaches

The reviewed study evaluated multiple machine learning models including Random Forest, Decision Tree, Support Vector Machine (SVM), Logistic Regression, and K-Nearest Neighbors (KNN) for crop recommendation. The models were trained using soil and agricultural datasets and assessed using evaluation metrics such as Accuracy, Precision, Recall, and F1-Score. The experimental results demonstrated that Random Forest outperformed several traditional approaches, achieving approximately 99% prediction accuracy. The study concluded that integrating machine learning with agricultural data can significantly improve decision-making, productivity, and sustainability in farming practices.

CHAPTER 3.

SOFTWARE REQUIREMENT SPECIFICATION

3.1 Introduction to SRS

The Software Requirements Specification (SRS) document provides a detailed description of the functional and non-functional requirements of the Crop Recommendation and Soil Health Analysis System. It serves as a blueprint for the design, development, testing of the application by clearly defining the system's features and behavior. The document outlines agricultural inputs such as nutrient measurements, temperature, humidity, soil pH, rainfall, and seasonal information, and describes how the system processes these inputs to generate crop recommendations, soil health assessments, and fertilizer suggestions. The SRS in the chapter is scoped as described in Chapter 1 and reflects the requirements as implemented at the time of this report.

3.2 Functional Requirements

1. The system shall allow users to register, log in, and securely access the platform. User credentials shall be validated before granting access to system functionalities.
2. The system shall allow users to manually enter agricultural parameters such as nutrient measurements (N, P, K), soil pH, temperature, humidity, rainfall, and season information for crop prediction
3. The system shall provide a geolocation feature that automatically detects the user's location and fetches environmental parameters such as temperature, humidity, and rainfall through external weather APIs, reducing manual data entry and improving prediction accuracy..
4. The system shall analyze the provided agricultural data using machine learning algorithms and generate the most suitable crop recommendation for the given conditions.
5. The system shall provide a ranked list of the top five most suitable crops based on prediction confidence and environmental compatibility
6. The system shall evaluate soil conditions using the provided nutrient and pH values and generate a quantitative soil health assessment.
7. The system shall recommend suitable fertilizers based on soil nutrient deficiencies and crop requirements to improve productivity and maintain soil fertility.
8. The system shall store user inputs, prediction results, and recommendation history in the database and allow users to access previous analyses when required.
9. The system shall communicate with external weather APIs to retrieve real-time environmental data based on the user's geographic location.
10. The system shall validate all user inputs and provide appropriate error messages when invalid, incomplete, or inconsistent data is submitted.

3.3 Non-Functional Requirements

- **Performance:** The system shall generate crop recommendations, soil health assessments, and fertilizer suggestions within a few seconds of receiving valid input data. API-based weather data retrieval should also be performed with minimal latency.
- **Reliability:** The system shall provide consistent and accurate recommendations based on the trained machine learning model and available agricultural data.
- **Usability:** The system shall provide an intuitive and user-friendly interface that enables farmers and agricultural stakeholders to use the application with minimal technical knowledge or training.
- **Portability:** The system shall run on simple hardware without requiring specialized or proprietary acceleration infrastructure.
- **Maintainability:** The application shall be designed using a modular architecture, making it easier to update, debug, test, and maintain individual components such as the frontend, backend, database, and machine learning modules.
- **Accuracy:** The machine learning model shall provide highly accurate crop recommendations based on the input parameters. The prediction accuracy should be continuously monitored and improved through model retraining and dataset updates.
- **Data Integrity :** The system shall validate all user inputs before processing and ensure that data stored in the database remains accurate, consistent.

3.4 Software Requirements

Table 1.1: Software Requirements

Component	Specification
Operating System	Windows 10/11, Linux or macOS
Programming Language	Python 3.11
Frontend Technologies	React.js,HTML5,CSS3,JavaScript
Backend Technologies	Node.js,Express.js
Machine Learning Technologies	Python 3.x,Scikit-Learn,Pandas,NumPy,Joblib
API Integration	Open-Meteo API(for weather data retrieval), Browser Geolocation API (for automatic location detection)
Data Handling	pandas, NumPy
Evaluation	scikit-learn (Evaluation metrics)

3.5 Hardware Requirements

Table 1.2: Hardware Requirements

Component	Minimum Specification Used
CPU	Intel Core i7 or equivalent
RAM	8 GB minimum (16 GB recommended)
Storage	10GB or more free disk space
Internet Connection	Internet for API integration,package installation
GPU	Not required

CHAPTER 4 UML DIAGRAMS

4.2.1 Use Case Diagram

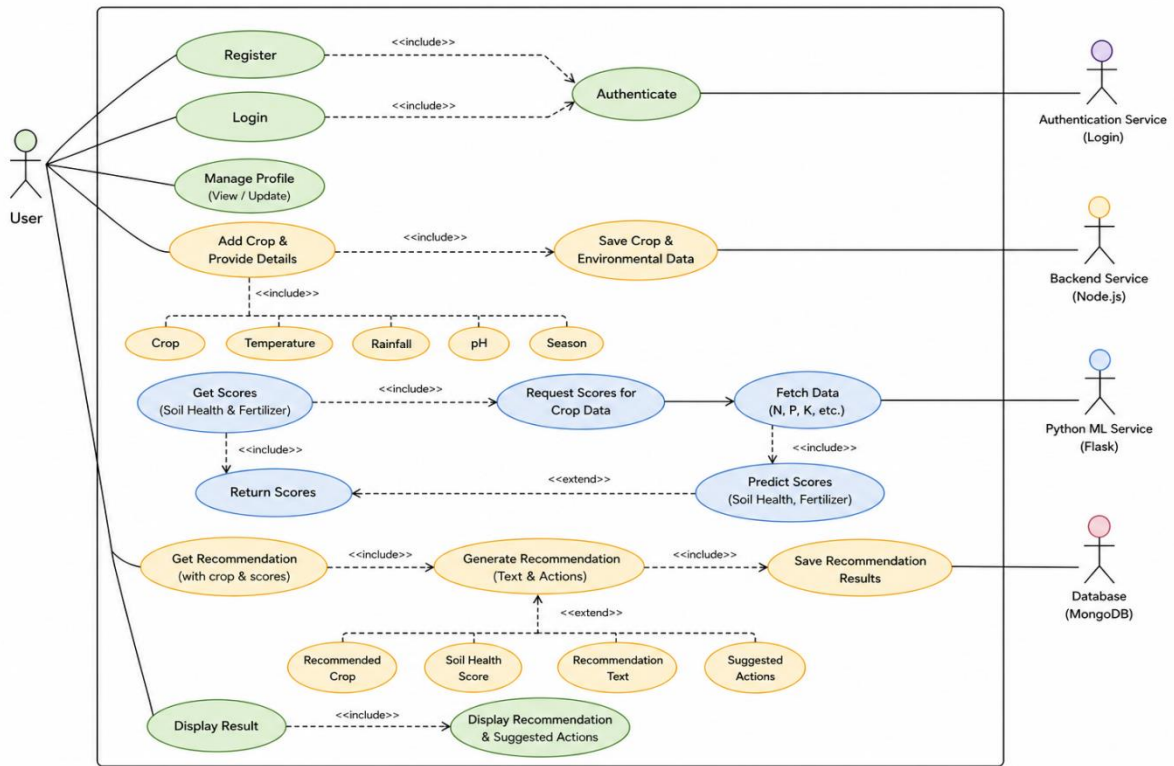


Fig 2.1: Use Case Diagram

The Use Case Diagram illustrates the interaction between the user and the various components of the Precision Agro Analytics system. The user can register and log into the platform, where authentication is handled through the authentication service and user provides agricultural information such as crop details, temperature, rainfall, soil pH, and seasonal data. The backend communicates with the Python-based machine learning service to fetch relevant soil and nutrient information and generate predictions. The ML model evaluates the input parameters and produces soil health scores and fertilizer recommendations. Based on these results, the system generates crop recommendations along with explanatory text and suggested actions. The recommendation results are then stored in the MongoDB database for future reference.

Finally, the user can view the generated recommendations, including the recommended crop, soil health score, fertilizer suggestions, and actionable insights through the system's user interface. This use case diagram serves as the highest level requirements reference for the entire system design.

4.2.2 Class Diagram

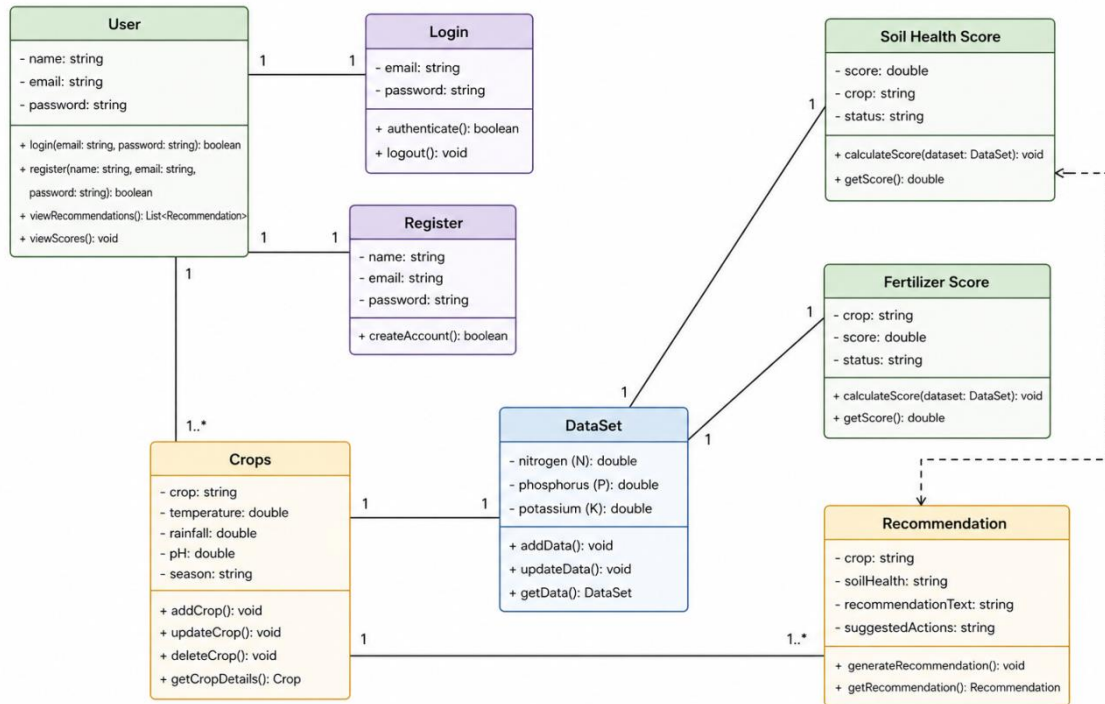


Fig 2.2: Class Diagram

The Class Diagram represents the structural design of the Precision Agro Analytics System by illustrating the relationships between the main classes and their functionalities. The User class is the central entity that stores user information such as name, email, and password and provides functionalities for login, registration, viewing recommendations, and accessing scores. The login and register classes are associated with the user class and handle authentication and account creation processes respectively.

The Crops class stores crop related information, including crop name, temperature, rainfall, soil pH, and season, and provides operations to add, update, delete, and retrieve crop details. The DataSet class acts as a repository for agricultural parameters such as Nitrogen (N), Phosphorus (P), and Potassium (K) values, which are used as input for prediction and analysis. The Soil Health Score and Fertilizer Score classes utilize the dataset to calculate soil health and fertilizer suitability scores based on the provided nutrient information. These calculated scores are then used by the Recommendation class to generate personalized crop recommendations, recommendation text, and suggested actions for the user. The relationships between the classes demonstrate the flow of information from user input to data processing and finally to recommendation generation, ensuring a structured and modular design for the system.

4.2.3 Sequence Diagram

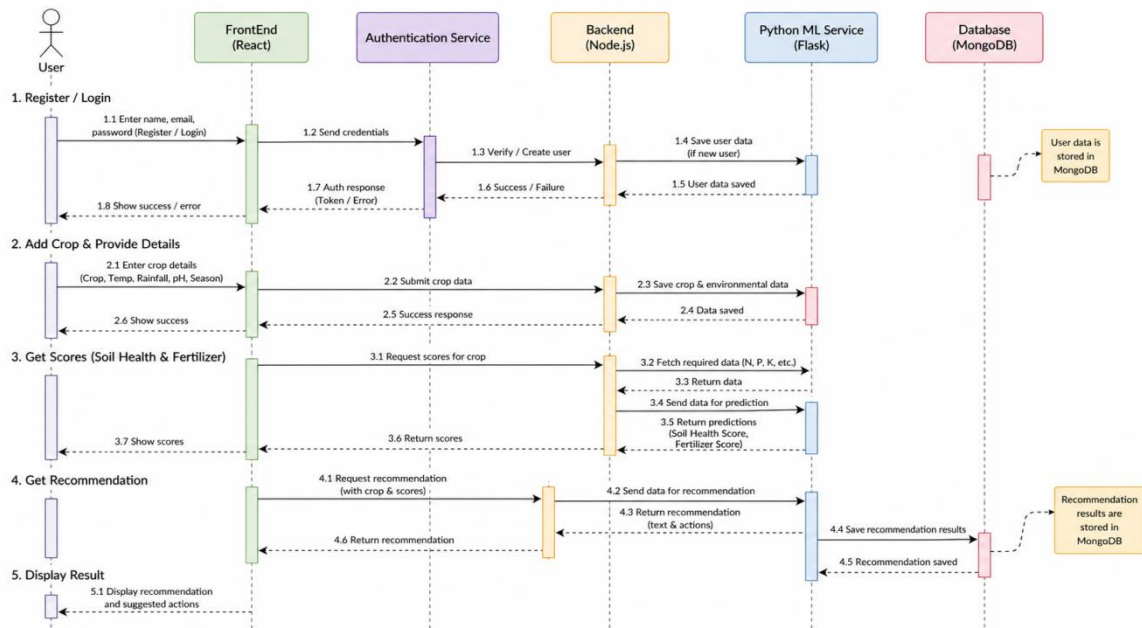


Fig 2.3: Sequence Diagram

The sequence diagram illustrates the interaction between the User, Frontend, Authentication Service, Backend, Python ML Service, and MongoDB Database. The process begins when the user registers or logs into the system through the frontend. The authentication service verifies the credentials and stores user information in the database if required. Then, the user enters crop and environmental details such as temperature, rainfall, soil pH, and season or can also fetch using the API's. The frontend sends this information to the Node.js backend, which stores the data and communicates with the Python Flask machine learning service. The ML service processes the input data and calculates soil health and fertilizer scores using the trained Random Forest model. Based on these scores, the system generates personalized crop recommendations and suggested actions. The recommendation results are then stored in MongoDB and returned to the frontend.

Finally, the user can view the recommended crop, soil health score, fertilizer suggestions, and other actionable insights through the dashboard.

4.2.4 Activity Diagram

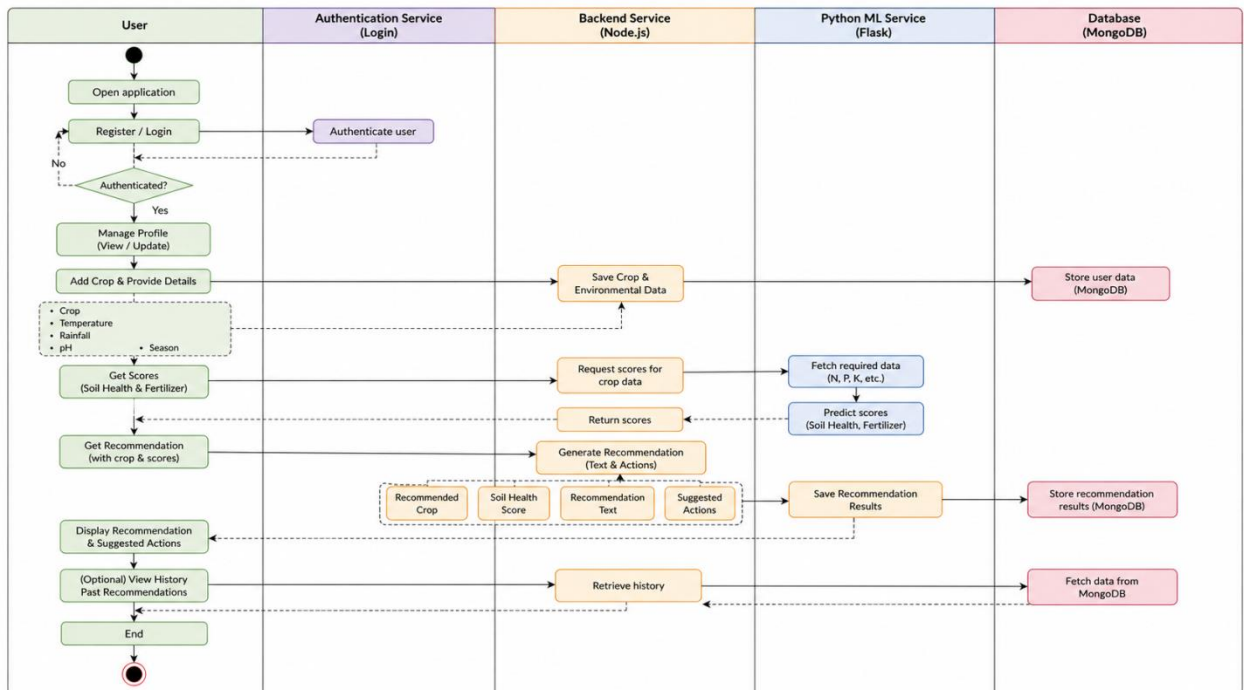


Fig 2.4: Activity Diagram

The activity diagram illustrates the complete workflow of the Precision Agro Analytics System from user authentication to recommendation generation. The process begins when the user opens the application and performs registration or login. After successful authentication, the user can manage their profile and provide agricultural information such as crop details, temperature, rainfall, soil pH, and season (or) can be fetched from API's. The backend service stores the submitted crop and environmental data in the MongoDB database. When the user requests soil health and fertilizer scores, the backend communicates with the Python Flask machine learning service, which retrieves the required data and generates prediction scores. These scores are returned to the backend and used to generate crop recommendations along with recommendation text and suggested actions. The generated recommendations are then stored in the database for future reference.

Finally, the system displays the recommended crop, soil health score, fertilizer recommendations, and suggested actions to the user. Additionally, the user can view the history of past recommendations before ending the session..

4.2.5 State Chart Diagram

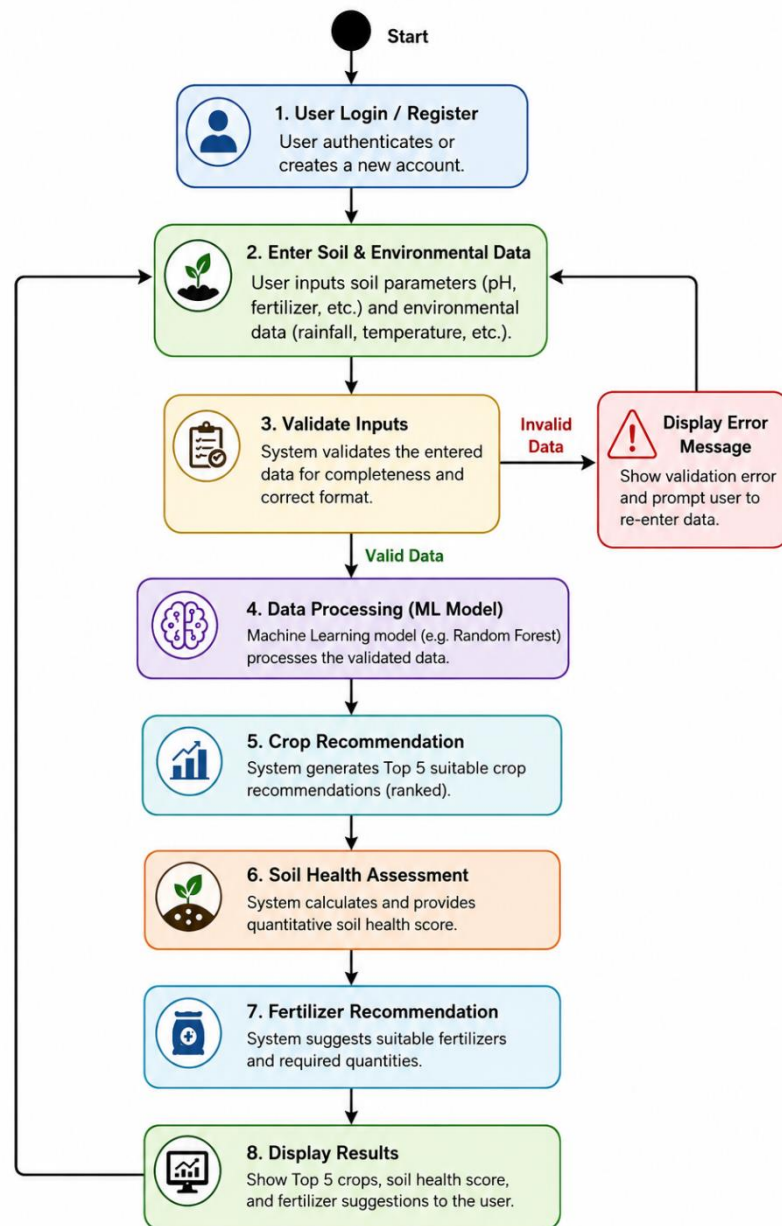


Fig 2.6: State Chart Diagram

The State Chart Diagram illustrates the workflow of the Precision Agro Analytics System. The process begins with user authentication, where the user logs in or registers to access the platform. The user then enters soil and environmental parameters such as pH, rainfall, temperature, and fertilizer values. The system validates the entered data and displays an error message if any invalid inputs are detected. Once the data is verified, the machine learning model processes the information to generate intelligent recommendations. Based on the analysis, the system provides the top five suitable crops, evaluates soil health, and suggests appropriate fertilizers. Finally, all recommendations and analysis results are displayed to the user, enabling informed decision making for sustainable and efficient farming practices.

4.2.6 Deployment Diagram

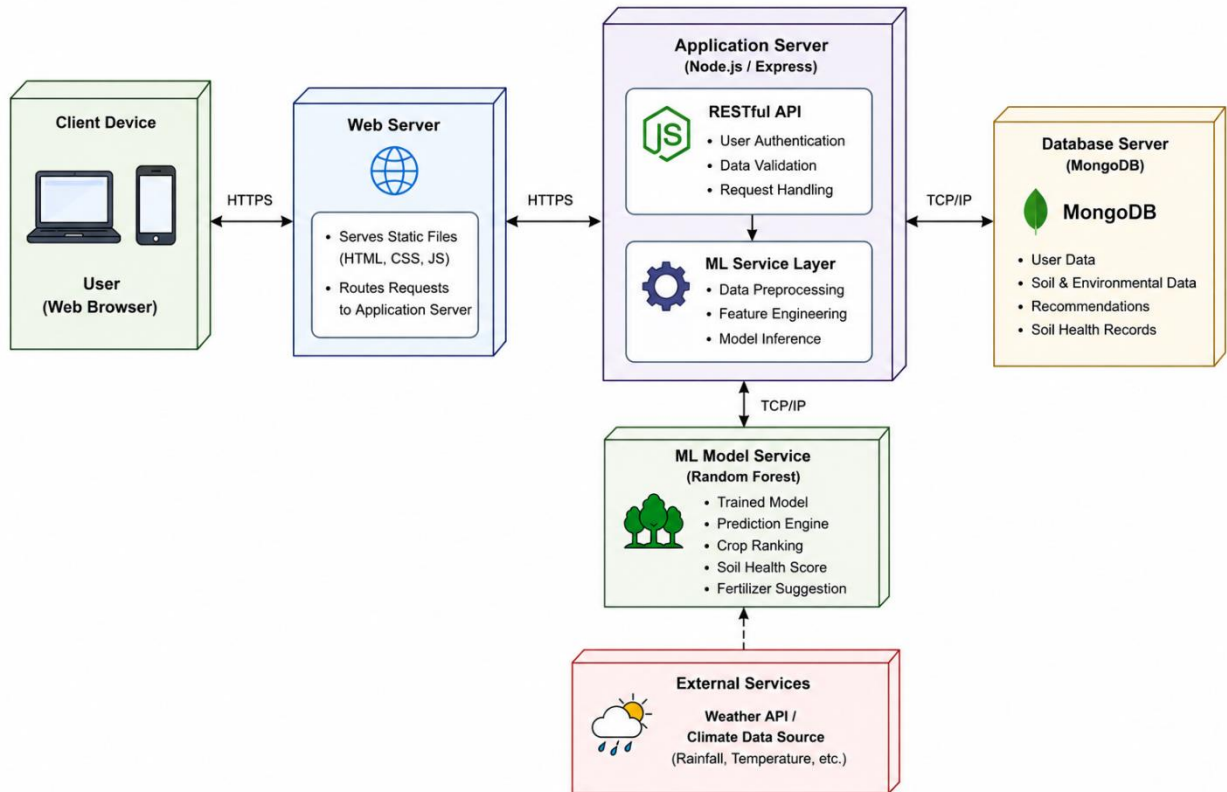


Fig 2.7: Deployment Diagram

The Deployment Diagram illustrates the physical architecture of the Smart Agriculture Crop and Soil Recommendation System. The user accesses the application through a web browser on a client device and communicates with the web server using HTTPS. The web server forwards requests to the application server, which handles user authentication, data validation, and request processing. The application server interacts with the MongoDB database to store and retrieve user, soil, and recommendation data. A dedicated Machine Learning Model Service processes agricultural parameters using the Random Forest algorithm to generate crop predictions, soil health assessments, and fertilizer recommendations. External weather and climate data services provide additional environmental information to improve prediction accuracy. This deployment architecture ensures secure communication, efficient data management, scalability, and real-time decision support for farmers.

CHAPTER 5 IMPLEMENTATION

5.1 Technologies and Sample Code

The implementation is focused on the development and integration of the Precision Agro Analytics System using the MERN stack and Machine Learning techniques. This phase involves designing the frontend interface, developing backend services, managing database operations, and integrating the Random Forest model for intelligent predictions. Users can input soil and environmental parameters or can fetch the inputs using the API's, which are processed by the system to generate crop recommendations, soil health analysis, and fertilizer suggestions. The implementation ensures efficient data flow, secure communication between system components, and real-time delivery of accurate agricultural insights.

Predictions

The system successfully processes the user provided parameters using the Random Forest machine learning model. Based on the prediction probabilities, the application identifies and ranks the top five most suitable crops along with their confidence scores. The highest ranked crop is displayed as the recommended crop, providing users with accurate and data driven decision support. Additionally, the results are presented in a user friendly format, enabling easy interpretation and effective agricultural planning.

```
predicted_crop = model.predict(features)[0]
probas         = model.predict_proba(features)[0]
top5_idx       = np.argsort(probas)[-5:][::-1]
top_five = [
    {
        'crop':      model.classes_[i],
        'confidence': round(float(probas[i]) * 100, 1)
    }
    for i in top5_idx
]

return jsonify({
    'topCrop':      predicted_crop,
    'confidence':   top_five[0]['confidence'],
    'topFive':      top_five,
    'season':       season_str
})
```


Feature Computation

The feature composition step combines all user inputs, including NPK values, temperature, humidity, pH, rainfall, and encoded season, into a single feature array that is used as input for crop prediction. Season encoding converts categorical season values such as Kharif, Rabi, and Zaid into numerical values (0, 1, and 2) so that the machine learning model can process them effectively.

```
SEASON_ENCODE = { 'kharif': 0, 'rabi': 1, 'zaid': 2 }
```

```
features = np.array([[
    float(data['nitrogen']),
    float(data['phosphorus']),
    float(data['potassium']),
    float(data['temperature']),
    float(data['humidity']),
    float(data['ph']),
    float(data['rainfall']),
    season_encoded
]])
```

Random Forest Model Training

The crop recommendation model was trained using the Crop_recommendation.csv dataset containing 11,566 samples across 22 different crop categories. A Random Forest Classifier with 200 decision trees was used to learn the relationship between soil nutrients (N, P, K), environmental factors (temperature, humidity, pH, rainfall), and crop suitability. The season information was encoded as numerical values and included as an additional feature to improve prediction accuracy. After training and evaluation, the model achieved an impressive 99% test accuracy, demonstrating its effectiveness in providing reliable crop recommendations. The trained model was integrated into the Flask API for real-time predictions.

```
Trained on : Crop_recommendation.csv (11,566 samples, 22 crops)
Algorithm   : Random Forest (200 trees, 99% test accuracy)
Features    : N, P, K, temperature, humidity, ph, rainfall, season_encoded
Season      : kharif=0, rabi=1, zaid=2
```

Crop Classification

The model analyzes eight input features: nitrogen, phosphorus, potassium, temperature, humidity, pH, rainfall, and season. Using the Random Forest algorithm, the model compares these inputs with patterns learned from 11,566 training samples covering 22 crop categories. For each prediction, the model calculates probability scores for all crops and ranks them according to their suitability. The crop with the highest confidence score is selected as the primary recommendation, while the remaining top ranked crops are presented as alternative options. This classification approach helps farmers make informed decisions by identifying the most suitable crops for their specific field conditions.

CHAPTER 6 RESULTS

6.1 Output, Results, and Reports

This section presents the outputs produced by the Precision Agro Analytics System, after processing user-provided soil and environmental parameters. The system was evaluated using various input combinations to assess the accuracy and effectiveness of crop prediction, soil health analysis, and fertilizer recommendation functionalities. By leveraging the Random Forest machine learning model, the application successfully generated reliable and data-driven recommendations. The obtained results demonstrate the system's ability to assist farmers in making informed agricultural decisions based on site specific conditions. These findings validate the effectiveness of integrating machine learning techniques into modern agricultural practices.

Feature Composition

The system successfully performs feature computation by transforming agricultural parameters into a structured format suitable for machine learning analysis. Inputs such as nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH, and rainfall are combined into a single feature array. Additionally, seasonal information is encoded into numerical values, where Kharif, Rabi and Zaid are represented as 0, 1 and 2 respectively. This encoding enables the Random Forest model to effectively interpret seasonal variations during prediction. The generated feature array serves as the final input to the model, ensuring accurate crop classification and recommendation based on the given field conditions.

Soil & Climate Inputs Enter values manually

Nitrogen (N)
60 mg/kg

Phosphorus (P)
40 mg/kg

Potassium (K)
40 mg/kg

Temperature
25 °C

Humidity
65 %

Soil pH
6.5

Rainfall
100 mm

Season
Kharif (Jun–Nov)

Predict Best Crop

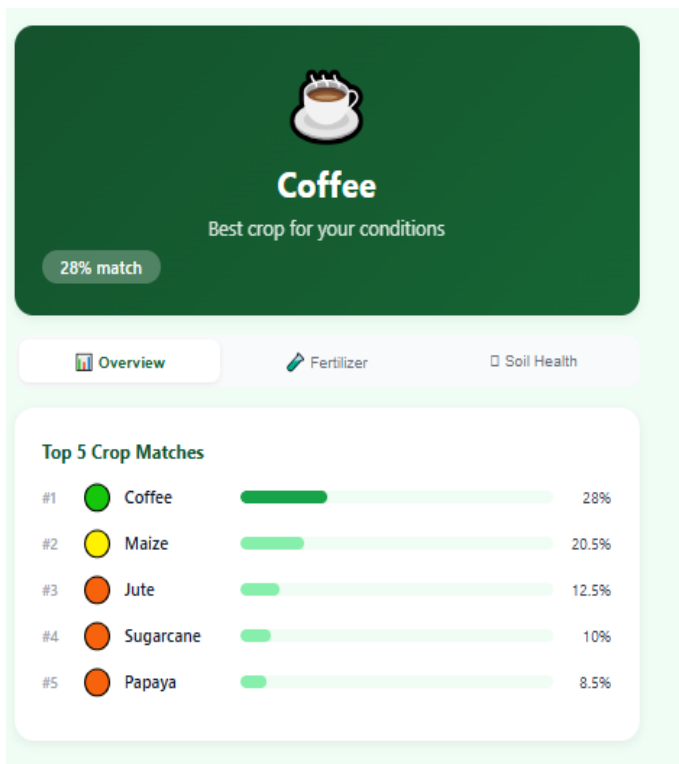
Input Parameters

Feature Statistics:									
	N	P	K	temperature	humidity	ph	rainfall	season_encoded	
count	11566.00	11566.00	11566.00	11566.00	11566.00	11566.00	11566.00	11566.00	
mean	56.64	52.20	47.14	24.84	70.09	6.51	103.28	0.53	
std	40.08	30.82	46.97	5.77	21.27	0.72	54.69	0.69	
min	0.00	5.00	5.00	8.03	14.26	3.50	20.21	0.00	

Feature Statistics

Predictions

After successfully processing the parameters, the Random Forest machine learning model analyzes the input features and calculates prediction probabilities for all supported crop categories. Based on these probabilities, the system identifies and ranks the top five most suitable crops along with their respective confidence scores. The crop with the highest confidence score is selected as the primary recommendation and displayed to the user. This approach provides accurate, reliable, and data-driven decision support for crop selection. Furthermore, the results are presented through an intuitive user interface, enabling easy interpretation and assisting farmers in effective agricultural planning and resource management.



CHAPTER 7 CONCLUSION

7.1 Summary of Project Findings

The Crop Recommendation and Soil Health Analysis System demonstrates the potential of machine learning and modern web technologies in addressing critical challenges faced by the agricultural sector. By integrating a Random Forest-based prediction model with a MERN-stack architecture, the system provides farmers with accurate, data-driven recommendations based on soil nutrients, temperature, rainfall, humidity, soil pH, and seasonal conditions. The platform not only identifies the most suitable crops and top five crop alternatives but also evaluates soil health and provides customized fertilizer recommendations, enabling more informed and efficient farming decisions.

Furthermore, the incorporation of geolocation-based weather data retrieval enhances the accuracy and convenience of the recommendation process by reducing manual data entry and providing real-time environmental information. The system promotes sustainable agricultural practices by optimizing resource utilization, improving crop productivity, and reducing the environmental impact of excessive fertilizer usage. Through its user-friendly interface and intelligent decision-support capabilities, the proposed solution bridges the gap between advanced data analytics and practical farming needs. Overall, the project highlights how machine learning-driven precision agriculture can contribute to increased agricultural productivity, farmer profitability, and long-term food security while paving the way for future enhancements such as market price prediction, crop disease detection, smart irrigation management, and farmer collaboration features.

In conclusion, the proposed system demonstrates how machine learning and modern web technologies can be effectively utilized to support precision agriculture and sustainable farming practices. By providing intelligent crop recommendations, soil health analysis, and fertilizer suggestions, the platform empowers farmers to make informed decisions that enhance productivity and resource efficiency. As agriculture continues to face challenges arising from climate change and increasing food demands, such data-driven solutions will play a vital role in building a more resilient, sustainable, and technology-enabled agricultural ecosystem.

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